Feasibility Study on Traditional Korean Desserts-Gaesong Juak (개성주악)

> This report briefly outlines
> This report presents a full data-driven feasibility study of the traditional Korean dessert market — from data collection and cleaning to analysis and business recommendations — using the potential launch of *Gaeseong Juak* (개성주악) in Seoul as a practical case study

Project Overview |

Core question	Evaluating the Market Feasibility of Traditional Korean Desserts Through Data, Not Intuition
Case study	This study is motivated by a real case — a friend's plan to open a *Gaeseong Juak* specialty shop in Seoul
Data	72 benchmark dessert shops in Seoul + 850+ consumer reviews across Korean dessert shops in Taipei, Tokyo, and Vancouver
Tools	Python、Apify、pandas、Selenium、matplotlib、js
Research period	March 2026-April 2026

What is *Gaeseong Juak*？ |

Gaeseong Juak has **a deep history**. Once reserved for royal banquets and high-status weddings, this treat is all about craftsmanship: **glutinous rice flour** fermented with Makgeolli, slow-fried to perfection, and finished with a glowing ginger-cinnamon glaze. It's famous for its "**crispy outside, chewy inside**" texture and an amber look that resembles a polished jewel.

What's truly exciting is its recent "modern makeover" in Seoul. Today's Gaeseong Juak has evolved **beyond its traditional roots, topped with everything from fresh fruits and premium nuts to artisanal creams.** This blend of ancient technique and modern aesthetics has made it an "Instagrammable" sensation. For the global market, it's more than just a sweet treat—it's a sophisticated cultural icon with massive potential to lead the premium dessert scene.

Research Approach |

This project adopts a human-AI collaborative workflow:

・**Anita Wu** — Research Design, Problem Statement, Analytical Logic, Code Review & Revision, Written Report

・**Claude AI（Anthropic)**-Code Generation,Data Processing

All key decisions were made by the researcher:

- Spotted and fixed a bias issue caused by shops that focus on one main item
- Switched from total count to **cross-shop occurrence rate** to better measure demand
- Built a three-dimensional scoring matrix with custom weights and definitions
- Scraped and cleaned the data multiple times until it was reliable
- Added existing Seoul shops selling Gaeseong Juak as a reference for future site selection

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The Framework |

This study developed a **four-step** product evaluation framework that can be reused for the market feasibility study of any food item

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Stage 1：Local Market Viability in Korea
  └ What does the competitive landscape look like, and what are the surviving       shops
selling？

Stage 2：Global Market Potential
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  └ What do consumers truly care about? Is it consistent across markets?

Stage 3：Cross-validation
  └ Does Gaesong Juak match consumer preferences ?

Stage 4：Site selection criteria
  └  Using existing Seoul shops that sell Gaeseong Juak as a basis for future site select
ion
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Stage 1：Local Market Viability in Korea ｜

Market Sentiment: Analyzing 72 Specialty Shops in Seoul

- **Consumer Familiarity:** High acceptance of core ingredients (Glutinous Rice, Red Bean, Black Sesame).
- **Benchmarking Success:** Successful business models already established by pioneers like **Yeonlihuijae** and **Hangwa Warak**.
- **Market Opportunity:** With only 25 shops currently selling it, the product sits in a "High Potential" category with low-to-medium competition.

Compiling a list of 72 benchmark dessert shops in Seoul

Research Design：Only shops with 3+ years of operation and 200+ reviews were included, ensuring the sample reflects **market-validated success** rather than passing trends.

Technical Challenges & Decisions: We started with Selenium to scrape Naver Map directly, but hit three major obstacles:

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Attempt 1: Direct API call → 404
The API token is dynamically generated and cannot be fixed, m
aking direct calls impossible
Attempt 2: Selenium browser simulation → Target DOM elements
not found
Naver Map's shop listings are rendered dynamically via JavaSc
ript. As a result, page_source only returns the skeleton HTML
— shop data is not included.
Diagnosis:

# Diagnosis:
driver.get("https://map.naver.com/p/search/전통디저트 홍대")
time.sleep(10)
```

```
html = driver.page_source
print(len(html))  # Only 78KB — confirmed shop data not yet r
endered
```

After three failed attempts, it was clear that Naver Map has a **robust anti-automation system.** Additionally, as a non-Korean researcher, accessing Naver data directly posed further barriers. The decision was made to switch to **Apify Naver Map Search Results Scraper** (https://console.apify.com/actors/UCpUxFUNcdKdbBdYg/input)as an alternative.

> 💡 Decision rationale：Time is a limited resource. Rather than continuing to fight anti-scraping mechanisms, finding a compliant and stable alternative was the smarter move — freeing up time for actual analysis. This decision saved an estimated **2–3 days** of unproductive engineering work.

Final Dataset Columns：

```
Each shop → Name,Category,Rating,Review Count,Address,Menu Items(with prices),
and Blog Reviews.
```

Item Scoring Matrix

Each menu item was scored across three dimensions (0–3 points each, 9 points total)

Dimension	Definition	Score Standard
Traditionality	Korean cultural authenticity	Historical ingredients · Traditional craftsmanship · Cultural symbolism
Standardizability	SOP scalability	Equipment requirements · Process complexity · Staff training difficulty
Longevity	3+ years of market data	Not a passing trend · Stable customer base · Consistent ingredient supply

Result：

Item	Traditionality	Standardizability	Longevity	Total
개성주악 (Gaeseong Juak)	⭐⭐⭐	⭐⭐	⭐⭐⭐	8/9
약과(Yakgwa)	⭐⭐⭐	⭐⭐⭐	⭐⭐⭐	9/9
빙수(Bingsu)	⭐⭐	⭐⭐	⭐⭐⭐	7/9

Item	Traditionality	Standardizability	Longevity	Total
식혜(Sikhye)	⭐⭐⭐	⭐⭐⭐	⭐⭐	8/9
소금빵(Salt Bread)	⭐	⭐⭐⭐	⭐⭐	6/9⚠️

> **Key Insight:** Gaeseong Juak scored lower than Yakgwa, but that's not the whole story. The gap comes from how easily each item can be standardized.
>
> Gaeseong Juak's complexity makes it harder to copy — that's a **competitive barrier**. Yakgwa's simplicity makes it easier to scale — that's a **scaling advantage**. Different strategies, not different values.

Analysis of Core Ingredients in the Seoul Dessert Market

Bridging Menu Trends and Consumer Preferences

1. Market Supply: Top Menu Ingredients (Menu Analysis)

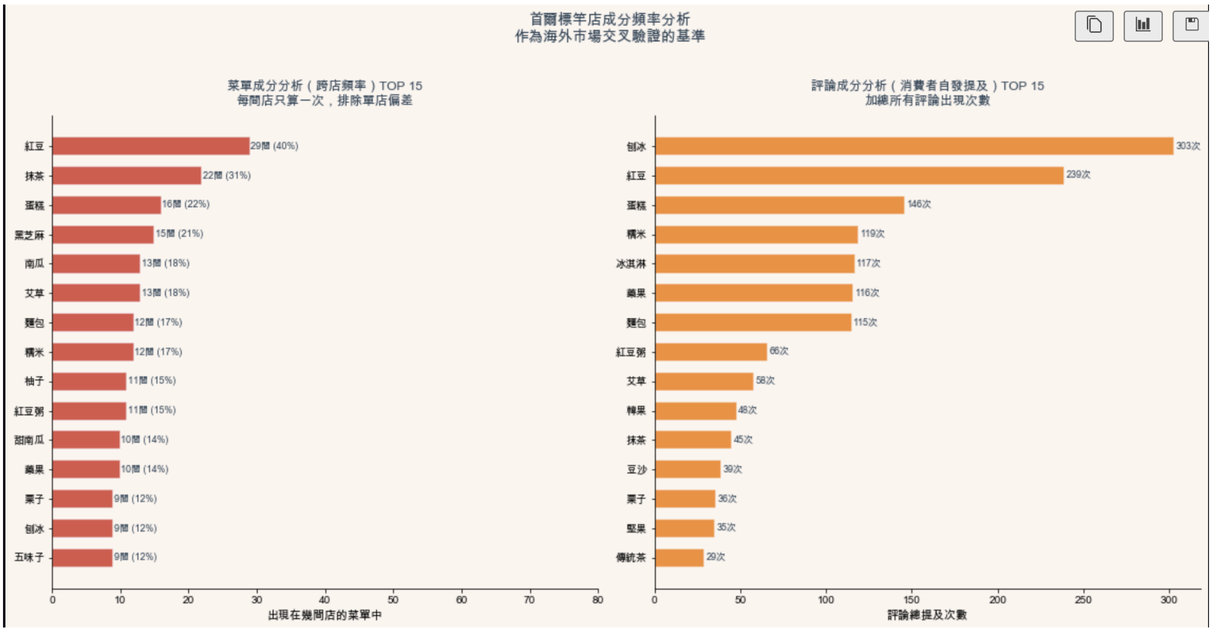
- **Dominance of Traditional Bases**: Red Bean (40%) and Matcha (31%) lead the menu frequency, establishing a solid foundation for traditional flavors in Seoul's premium dessert shops.
- **The Rise of Health-Conscious Flavors**: Traditional healthy ingredients like Black Sesame (21%), Pumpkin (18%), and Mugwort (18%) show high penetration across multiple locations, indicating a mature market for "K-Wellness" desserts.
- **Structural Ingredients**: Glutinous Rice appears in 17% of surveyed menus, serving as a critical texture-defining component for traditional pastries.

2. Consumer Demand: High-Frequency Keywords in Reviews (Review Sentiment Analysis)

- **Texture & Experience**: "Shaved Ice" (303 mentions) and "Glutinous Rice" (119 mentions) are among the top keywords, highlighting that consumers prioritize unique textures (chewiness and coolness).
- **Ingredient Resonance**: Red Bean remains the most discussed specific ingredient (239 mentions), proving that it is not only widely available but also highly valued by customers.
- **Emerging Interest in Heritage**: "Yakgwa" (116 mentions) and "K-Sweets" (48 mentions) show that traditional confectionery is a significant topic of spontaneous consumer discussion.

3. Strategic Validation for Gaeseong Juak (Market Opportunity)

- **Alignment with "Proven" Success**: Gaeseong Juak is primarily composed of **Glutinous Rice** and often paired with **Red Bean** or **Nuts**. Data shows these are the exact ingredients with the highest overlap between menu availability and consumer mentions.
- **Market Gap Exploitation**: While ingredients like Shaved Ice are seasonal, Glutinous Rice-based snacks offer year-round appeal and satisfy the consumer's documented desire for traditional, "not too sweet" textures.



Stage 2：Global Market Potential

Research Design

Do Global Consumers Align with the Texture and Heritage of Gaeseong Juak?：

This study analyzes 850 consumer reviews across Taipei, Tokyo, and Vancouver to find out whether global consumers naturally resonate with what 개성주악 has to offer.

Data collect：Apify Google Maps Scraper，Taipei-5 shops, 250 reviews、Tokyo-6 shops 300 reviews、Vancouver 6 shops 300 reviews，total 850 reviews.

Methodological Refinement: Correcting Supply-Side Bias

Initial Approach & Limitation：Initially, the analysis relied on a simple **aggregate frequency count** of keywords across all reviews. This led to a significant distortion where "**Yakgwa**" was ranked disproportionately high. Upon deeper investigation, this spike was attributed to a single high-volume store—**ROZI coffee** in Tokyo—where the menu specifically features Yakgwa.

```
# ❌ Initial approach：aggregate frequency count of keywords across all reviews
total = sum(review.count("약과") for review in all_reviews)

# ✅ Revise：Cross-Store Frequency Filtering
shop_counts = {}
for shop in df["店名"].unique():
    shop_text = " ".join(df[df["店名"]==shop]["評論"].tolist())
    count = sum(shop_text.count(w) for w in keywords["藥果"])
    if count > 0:
        shop_counts[shop] = count

cross_shop_num = len(shop_counts)
```

Key Methodological Correction

To ensure the analysis accurately reflects true consumer intent, the following refinement was implemented:

1. **Cross-Store Frequency Filtering :**
   Instead of raw counts, ingredients are now ranked based on the **number of unique establishments** where they were mentioned by consumers. This filters out "single-store outliers" and identifies trends that are consistent across the entire market.
2. **Resulting Insight:**
   By neutralizing the ROZI coffee bias, **"Glutinous Rice "** and **"Balanced Sweetness "** emerged as the true universal drivers of satisfaction across Taipei, Tokyo, and Vancouver. This provides a much more robust validation for **Gaeseong Juak**, as its core appeal aligns with these broad market demands rather than a localized trend.

Results

Data Interpretation Guide: Identifying Market-Wide Trends:

- **Bias Exclusion**: Keywords mentioned in only a single store were excluded to prevent "Single-Store Effects" from distorting the results.

- **Validation Criterion**: Only attributes appearing in **2+ unique locations** were counted as valid indicators of general consumer preference.
- **Goal**: To define the ingredients and flavors that resonate most with the **general public** across different regions.

Keywords	Taipei	Tokyo	Vancouver	result
nuts	5 shops • 14 times	2 shops • 30 times	2 shops • 50 times	🔴 Strongest Signals Across Three Cities
balanced sweetness	5 shops • 24 times	4 shops • 13 times	5 shops • 8 times	🔴 Universal Preferences in All Three Cities
crispy, crunch	4 shops • 24 times	4 shops • 12 times	4 shops • 5 times	🔴 Cross-Border Texture Demands
Korea-Aesthetics	5 shops • 46 times	6 shops • 96 times	4 shops • 25 times	🔴 Strong Cultural Resonance / Authenticity
Repurchase	5 shops • 58 times	6 shops • 37 times	5 shops • 19 times	🔴 High Customer Retention
Gaeseong Juak	✗ Untapped	✗ Untapped	✗ Untapped	⚪ **Zero Overseas Brand Awareness**

According to findings identify a 'Triple-Threat' preference **(Nutty, Balanced Sweetness, Crispy)** common across global markets. Gaeseong Juak naturally aligns with these demands. Given the current **Zero Overseas Brand Awareness**, early market entry offers a unique opportunity to shape the category's brand identity

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Stage 3：Cross-validation ｜

Do local market foundations and overseas demands collectively support the viability of Gaeseong Juak?

Validation Dimension	Local Market Data (Seoul)	Overseas Market Data (Global)	Product Alignment: Gaeseong Juak
Traditional Ingredient Base	Glutinous Rice has a proven foundation	Strong preference for Authentic Korean	✅ Perfect Match

Validation Dimension	Local Market Data (Seoul)	Overseas Market Data (Global)	Product Alignment: Gaeseong Juak
	across 72 shops.	Essence and heritage.	
Texture Demand	Traditional deep-frying techniques are widely established.	Consistent preference for Crispy textures across all three cities.	✅ Inherent Attribute
Sweetness Profile	Honey-based sweetness is a hallmark of traditional methods.	Universal preference for Balanced Sweetness (Not too sweet).	✅ Inherent Attribute
Nutty Elements	few presence/visibility in the current Seoul market.	Strongest consumer signal (58 mentions) across three cities.	✅ Core Strength (Traditional recipes feature nut fillings)
Market Awareness	Successful business models already established in Seoul.	Completely unmanifested in overseas markets (Zero awareness).	⚠️ First-Mover Opportunity

Core Conclusion：Data from 72 Seoul shops confirms the traditional ingredient foundation; 850 overseas reviews validate consumer taste preferences. Two independent datasets point to the same conclusion — **Gaeseong Juak's ingredients and texture naturally align with market demand.**

Stage 4：Site selection criteria |

If Viable, Where Should It Open?

Core Objective: To identify Seoul districts that offer a high density of target customers while minimizing direct competition.

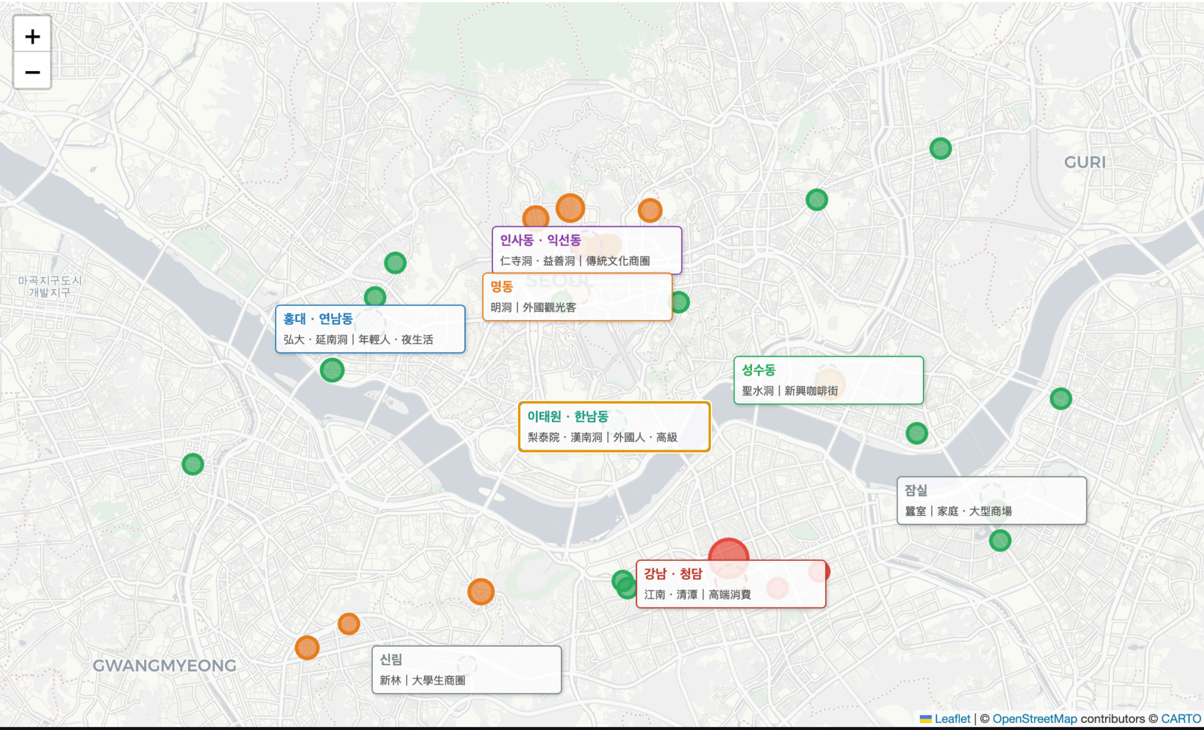
Competitive Intensity Matrix (Metrics)

Status	Condition	Strategic Implication
🔴 **High**	Max Reviews ≥3000	Market is dominated by established, high-traffic players.
🟡 **Medium**	Max Reviews ≥500 OR Total Shops ≥ 3	Competition exists but remains open to differentiation.
🟢 **Low**	Max Reviews < 500 AND Total Shops ≤2	Significant entry window with lower barrier to entry

Note on Limitations: This intensity model is primarily based on review volume and does not yet account for brand positioning overlap or micro-growth trends within neighborhoods.

Seoul District Competitive Landscape

District	Intensity	Key Competitors	Max Review Count	Recommendation
Gangnam-gu	🔴 High	Hangwa Warak	5,679	Avoid
Seongdong-gu	🟡 Med	Yebindang	1,950	Differentiated Entry
Jongno-gu	🟡 Med	Yeongyeongdang, Jjuak Jjuak	1,257	Differentiated Entry
Gwanak-gu	🟡 Med	Seoul Butter Sand	672	Monitor
Mapo-gu	🟢 Low	Myeonggwajeong	372	High Opportunity
Jung-gu	🟢 Low	Milgwabang	144	High Opportunity
Others	🟢 Low	Single boutique shops	20 ~ 192	High Opportunity



Strategic Site Recommendations

- **Primary Choice: Jongno-gu (Insadong / Ikseon-dong)**

    - **Rationale:** As a traditional cultural hub, the existing customer base has a pre-established affinity for Korean heritage desserts. The cultural atmosphere perfectly aligns with the premium positioning of Gaeseong Juak. While competition is moderate, success can be achieved through **brand storytelling and superior product aesthetics.**
- **Secondary Choice: Jung-gu (Myeong-dong & Surrounding areas)**
    - **Rationale:** These areas are dense with international tourists. For this demographic, Gaeseong Juak offers a "Novelty Factor." With low direct competition, this location provides a strong **First-Mover Advantage** to define the product for a global audience.
- **Areas to Avoid: Gangnam-gu**
    - **Rationale:** Dominated by a strong flagship brand (Hangwa Warak). The market here is nearing saturation for this specific niche, making the **cost of customer acquisition and entry prohibitively high.**

Business Recommendations

Prioritizing Seoul: Domestic Stability Over Early Overseas Expansion

- **The Challenge of Consumer Awareness:** Overseas data explicitly shows that consumer awareness for **Gaeseong Juak** is currently near zero. The "Market Education Cost" required for international expansion at this stage would significantly outweigh the potential ROI.
- **Proven Domestic Success:** The viability of the Seoul market has already been validated by successful pioneers like **Yeonlihuijae** (over 2,793 reviews), suggesting that focusing on the domestic base is the safer, more strategic first step.

Site Selection: Leveraging Cultural Density

- **Targeting High-Affinity Districts:** The primary locations should be **Insadong, Bukchon, or Ikseon-dong**—districts with a high "Cultural Density".
- **Reducing "Cold Start" Costs:** Consumers in these traditional hubs already have a high acceptance rate for Korean heritage desserts. By opening here, the brand can leverage existing cultural interest rather than educating the market from scratch, drastically lowering initial marketing expenses.

Menu Strategy: "Yakgwa" as the Entry, "Gaeseong Juak" as the future

- **Entry-Level Item (Yakgwa):** Given that **Yakgwa** currently has higher global recognition and market validation, it should serve as the "Entry Point" product to attract customers who are already familiar with the Korean culture.
- **The Core Differentiator (Gaeseong Juak):** Once customers are drawn in by the familiarity of Yakgwa, **Gaeseong Juak** functions as the "Differentiator" and "Competitive Moat." Its unique texture and aesthetic appeal will be the key to customer retention and establishing long-term brand loyalty.

Limitations |

While this analysis provides a strategic framework for market entry, the following limitations should be considered when interpreting the results:

- **Sample Size Constraints** : With only 5–6 representative shops analyzed per overseas city, the findings indicate a high-level direction but may lack granular precision.
- **Sampling Bias** : Data retrieved via Apify may not capture all flagship or hidden gem locations, potentially leading to a selection bias in the overseas dataset.
- **The "Zero Awareness" Caveat** : The absence of **Gaeseong Juak** in overseas reviews may reflect a gap in data coverage rather than a complete lack of market demand or existence.
- **Competitiveness Metrics** : The assessment of competitive intensity is primarily based on review volume; it does not fully account for brand positioning overlaps or emerging micro-trends.
- **Search Engine Constraints**: The identification of 25 competitors in Seoul was subject to Naver Map’s ranking algorithms, which may have omitted certain niche or newly opened boutiques.

Tech Stack |

```
Data Collection
├── Apify Naver Map Search Results Scraper
│     Seoul 72 shops・menu items・blog reviews
├── Apify Google Maps Scraper
│     Taipei 6 shops・Tokyo 6 shops・Vancouver 6 shops
└── Selenium + ChromeDriver
      Attempted direct scraping via Selenium — switched to Apify after encountering anti-b
ot mechanisms
```

```
Data Processing
├── Python 3.9
├── pandas — data cleaning and wrangling
├── collections.Counter — - Keyword frequency analysis — cross-shop method to eliminate si
ngle-store bias
└── json — Structured data parsing

Analytical Methods
├── Cross-shop Keyword Frequency
│     Eliminates single-store supply bias from demand signals
├── Multi-dimensional Scoring Matrix
│     Authenticity · Scalability · Market Longevity
└── Cross-market Validation
      Consumer demand comparison: Taipei · Tokyo · Vancouver

Visualization
├── matplotlib · seaborn — Chart generation (10 charts)
└── Folium — Interactive Seoul competitor map with district labels

Output
├── Jupyter Notebook (full analysis with charts)
├── GitHub Repository
└── Notion Portfolio (this document)
```

What I Learned

1. When blocked, reroute — don't force through

Naver Map's anti-bot mechanisms cost me significant time. Switching to Apify was the right call. Engineering decisions aren't just technical — they're about allocating time wisely.

2. Catching your own bias mid-research is more valuable than getting it right the first time

The cross-shop keyword correction came from noticing data distortion halfway through. A good analyst isn't someone who never makes mistakes — it's someone who catches and corrects them before drawing conclusions.

3. The right framework matters more than more data

Reframing from "supply-side vs demand-side" to "local acceptance → overseas acceptance → cross-validation → location" made the entire research narrative immediately clearer and more intuitive.

4. Honest data beats a clean conclusion

Multiple times throughout this project, the data didn't support the narrative I was building. Each time, I chose to revise the conclusion rather than force the numbers to fit.

Anita Wu • 2026 .04 Python • Apify • Claude AI